

7(a)(i)	<i>oxygenated</i> 1 percentage (of root tissue as aerenchyma), remains low ; any two from: <i>deoxygenated</i> 2 percentage (of root tissue as aerenchyma), increases gradually / small increase, initially / until 24 hours ; 3 percentage (of root tissue as aerenchyma), larger / steeper / more rapid, increase after 24 hours ; 4 percentage (of root tissue as aerenchyma), stays the same / is constant / plateaus / levels off / reaches maximum, at, the end / 48 hours ; 5 data quote ; <i>two times and two percentages with units</i> <table><tr><th>time / h</th><th>percentage of root tissue as aerenchyma when deoxygenated</th></tr><tr><td>0</td><td>0.1 – 0.3</td></tr><tr><td>12</td><td>0.5 – 0.6</td></tr><tr><td>24</td><td>0.8</td></tr><tr><td>36</td><td>2.5 – 2.6</td></tr><tr><td>48</td><td>6.5 – 6.6</td></tr><tr><td>60</td><td>6.5 – 6.6</td></tr></table>	time / h	percentage of root tissue as aerenchyma when deoxygenated	0	0.1 – 0.3	12	0.5 – 0.6	24	0.8	36	2.5 – 2.6	48	6.5 – 6.6	60	6.5 – 6.6	3
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7(a)(ii)	1 (aerenchyma is a tissue) with many / more / large, air spaces ; 2 (so) <u>oxygen</u>, diffuses / moves, (from the aerial parts) to root (cells) / lower parts of the plant / AW ; 3 (for) <u>aerobic</u> respiration ;	3														

Question	Answer	Marks
7(b)	any four from: 1 fast(er), stem / internode, growth ; 2 (so) leaves / flowers, above water level ; 3 (so) photosynthesis / gas exchange / reproduction, can occur ; 4 (root cells) tolerant to (increased concentration of) ethanol or (root cells) have more ethanol dehydrogenase ; 5 so, anaerobic respiration / ethanol fermentation, can occur ; 6 AVP ; e.g. leaves have ridges 7 AVP ; e.g. to trap air (so oxygen can be taken in by plant)	4